RYU Task1

Creating topology of linear 2Switches 2hosts per switches.

```
mininet@mininet-vm:~$ sudo mn --topo linear,2,2
*** Creating network
*** Adding controller
*** Adding hosts:
h1s1 h1s2 h2s1 h2s2
*** Adding switches:
s1 s2
*** Adding links:
(h1s1, s1) (h1s2, s2) (h2s1, s1) (h2s2, s2) (s2, s1)
*** Configuring hosts
h1s1 h1s2 h2s1 h2s2
*** Starting controller
c0
*** Starting 2 switches
s1 s2 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1s1 -> h1s2 h2s1 h2s2
h1s2 -> h1s1 h2s1 h2s2
h2s1 -> h1s1 h1s2 h2s2
h2s2 -> h1s1 h1s2 h2s1
*** Results: 0% dropped (12/12 received)
mininet> h1s1 ping h1s2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.229 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.055 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.087 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.090 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.072 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.076 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.072 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.072 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.103 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.150 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.076 ms
^C
--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19447ms
rtt min/avg/max/mdev = 0.046/0.082/0.229/0.040 ms
mininet> _
```

2. 3Switches 2Hosts per switch

```
mininet@mininet-vm:~$ sudo mn --topo linear,3,2
*** Creating network
*** Adding controller
*** Adding hosts:
h1s1 h1s2 h1s3 h2s1 h2s2 h2s3
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1s1, s1) (h1s2, s2) (h1s3, s3) (h2s1, s1) (h2s2, s2) (h2s3, s3) (s2, s1) (s3, s2)
*** Configuring hosts
h1s1 h1s2 h1s3 h2s1 h2s2 h2s3
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1s1 -> h1s2 h1s3 h2s1 h2s2 h2s3
h1s2 -> h1s1 h1s3 h2s1 h2s2 h2s3
h1s3 -> h1s1 h1s2 h2s1 h2s2 h2s3
h2s1 -> h1s1 h1s2 h1s3 h2s2 h2s3
h2s2 -> h1s1 h1s2 h1s3 h2s1 h2s3
h2s3 -> h1s1 h1s2 h1s3 h2s1 h2s2
*** Results: 0% dropped (30/30 received)
```

3. 3Switches 1 Host per Switches

```
mininet@mininet-vm:~$ sudo mn --topo linear,3,1
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1, s1) (h2, s2) (h3, s3) (s2, s1) (s3, s2)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

4. Single Switch 3 hosts

```
mininet@mininet-vm:~$ sudo mn --topo single,3
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

5. RYU installation and running 3 host 1 switch setup

```
Last login: Mon Jan  5 15:29:48 PST 2026 on tty1
mininet@mininet-vm:~$ sudo mn --topo linear,2,2 --controller=remote,ip=127.0.0.1
*** Creating network
*** Adding controller
Connecting to remote controller at 127.0.0.1:6653
*** Adding hosts:
h1s1 h1s2 h2s1 h2s2
*** Adding switches:
s1 s2
*** Adding links:
(h1s1, s1) (h1s2, s2) (h2s1, s1) (h2s2, s2) (s2, s1)
*** Configuring hosts
h1s1 h1s2 h2s1 h2s2
*** Starting controller
c0
*** Starting 2 switches
s1 s2 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1s1 -> h1s2 h2s1 h2s2
h1s2 -> h1s1 h2s1 h2s2
h2s1 -> h1s1 h1s2 h2s2
h2s2 -> h1s1 h1s2 h2s1
*** Results: 0% dropped (12/12 received)
mininet> _
```

The Ryu Controller running in Other Terminal.

```
^A^Cmininet@mininet-vm:~$ ryu-manager ryu.app.simple_switch_13
loading app ryu.app.simple_switch_13
loading app ryu.controller.ofp_handler
instantiating app ryu.app.simple_switch_13 of SimpleSwitch13
instantiating app ryu.controller.ofp_handler of OFPHandler
packet in 1 22:03:1e:91:73:f6 ff:ff:ff:ff:ff:ff 1
packet in 2 22:03:1e:91:73:f6 ff:ff:ff:ff:ff:ff 3
packet in 2 42:61:55:46:f2:6d 22:03:1e:91:73:f6 1
packet in 1 42:61:55:46:f2:6d 22:03:1e:91:73:f6 3
packet in 1 22:03:1e:91:73:f6 42:61:55:46:f2:6d 1
packet in 2 22:03:1e:91:73:f6 42:61:55:46:f2:6d 3
packet in 1 22:03:1e:91:73:f6 ff:ff:ff:ff:ff:ff 1
packet in 2 22:03:1e:91:73:f6 ff:ff:ff:ff:ff:ff 3
packet in 1 96:55:89:0a:20:bc 22:03:1e:91:73:f6 2
packet in 1 22:03:1e:91:73:f6 96:55:89:0a:20:bc 1
packet in 1 22:03:1e:91:73:f6 ff:ff:ff:ff:ff:ff 1
packet in 2 22:03:1e:91:73:f6 ff:ff:ff:ff:ff:ff 3
packet in 2 f6:19:37:c3:7a:eb 22:03:1e:91:73:f6 2
packet in 1 f6:19:37:c3:7a:eb 22:03:1e:91:73:f6 3
packet in 1 22:03:1e:91:73:f6 f6:19:37:c3:7a:eb 1
packet in 2 22:03:1e:91:73:f6 f6:19:37:c3:7a:eb 3
packet in 2 42:61:55:46:f2:6d ff:ff:ff:ff:ff:ff 1
packet in 1 42:61:55:46:f2:6d ff:ff:ff:ff:ff:ff 3
packet in 1 96:55:89:0a:20:bc 42:61:55:46:f2:6d 2
packet in 2 96:55:89:0a:20:bc 42:61:55:46:f2:6d 3
packet in 2 42:61:55:46:f2:6d 96:55:89:0a:20:bc 1
packet in 1 42:61:55:46:f2:6d 96:55:89:0a:20:bc 3
packet in 2 42:61:55:46:f2:6d ff:ff:ff:ff:ff:ff 1
packet in 2 f6:19:37:c3:7a:eb 42:61:55:46:f2:6d 2
packet in 1 42:61:55:46:f2:6d ff:ff:ff:ff:ff:ff 3
packet in 2 42:61:55:46:f2:6d f6:19:37:c3:7a:eb 1
packet in 1 96:55:89:0a:20:bc ff:ff:ff:ff:ff:ff 2
packet in 2 96:55:89:0a:20:bc ff:ff:ff:ff:ff:ff 3
packet in 2 f6:19:37:c3:7a:eb 96:55:89:0a:20:bc 2
packet in 1 f6:19:37:c3:7a:eb 96:55:89:0a:20:bc 3
packet in 1 96:55:89:0a:20:bc f6:19:37:c3:7a:eb 2
packet in 2 96:55:89:0a:20:bc f6:19:37:c3:7a:eb 3
```

3 Switches ,1 Hosts Each

```
mininet@mininet-vm:~$ sudo mn --topo linear,3,1 --controller=remote,ip=127.0.0.1
*** Creating network
*** Adding controller
Connecting to remote controller at 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1 s2 s3
*** Adding links:
(h1, s1) (h2, s2) (h3, s3) (s2, s1) (s3, s2)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

Tree Topology – Depth 2, Fanout 2

```
mininet@mininet-vm:~$ sudo mn --topo tree,2,2 --controller=remote,ip=127.0.0.1
*** Creating network
*** Adding controller
Connecting to remote controller at 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1 s2 s3
*** Adding links:
(s1, s2) (s1, s3) (s2, h1) (s2, h2) (s3, h3) (s3, h4)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
```

Linear – 4 switches, 2 hosts each

```
completed in 314.125 seconds
mininet@mininet-vm:~$ sudo mn --topo linear,4,2 --controller=remote,ip=127.0.0.1,port=6653
*** Creating network
*** Adding controller
*** Adding hosts:
h1s1 h1s2 h1s3 h1s4 h2s1 h2s2 h2s3 h2s4
*** Adding switches:
s1 s2 s3 s4
*** Adding links:
(h1s1, s1) (h1s2, s2) (h1s3, s3) (h1s4, s4) (h2s1, s1) (h2s2, s2) (h2s3, s3) (h2s4, s4) (s2, s1) (s3, s2) (s4, s
*** Configuring hosts
h1s1 h1s2 h1s3 h1s4 h2s1 h2s2 h2s3 h2s4
*** Starting controller
c0
*** Starting 4 switches
s1 s2 s3 s4 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1s1 -> h1s2 h1s3 h1s4 h2s1 h2s2 h2s3 h2s4
h1s2 -> h1s1 h1s3 h1s4 h2s1 h2s2 h2s3 h2s4
h1s3 -> h1s1 h1s2 h1s4 h2s1 h2s2 h2s3 h2s4
h1s4 -> h1s1 h1s2 h1s3 h2s1 h2s2 h2s3 h2s4
h2s1 -> h1s1 h1s2 h1s3 h1s4 h2s2 h2s3 h2s4
h2s2 -> h1s1 h1s2 h1s3 h1s4 h2s1 h2s3 h2s4
h2s3 -> h1s1 h1s2 h1s3 h1s4 h2s1 h2s2 h2s4
h2s4 -> h1s1 h1s2 h1s3 h1s4 h2s1 h2s2 h2s3
*** Results: 0% dropped (56/56 received)
```

Using tcpdump to Monitor Network Flows in Mininet

Topology Used (Minimum 2 Switches) #sudo mn --topo linear,2,1

```
*** Adding hosts:
h1 h2
*** Adding switches:
s1 s2
*** Adding links:
(h1, s1) (h2, s2) (s2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 2 switches
s1 s2 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s2-eth1
s1 lo:  s1-eth1:h1-eth0 s1-eth2:s2-eth2
s2 lo:  s2-eth1:h2-eth0 s2-eth2:s1-eth2
c0
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.672 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.155 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.112 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.072 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.150 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.168 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.136 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.178 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.171 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.107 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.155 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.108 ms
^C
--- 10.0.0.2 ping statistics ---
21 packets transmitted, 21 received, 0% packet loss, time 20466ms
rtt min/avg/max/mdev = 0.049/0.136/0.672/0.126 ms
mininet>
```

Run tcpdump on a switch interface

Example: Capture traffic on switch s1

```
16:28:18.350252 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2269, seq 1, length 64
16:28:18.354135 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2269, seq 1, length 64
16:28:18.376276 IP 10.0.0.2 > 10.0.0.1: ICMP echo request, id 2271, seq 1, length 64
16:28:18.376830 IP 10.0.0.1 > 10.0.0.2: ICMP echo reply, id 2271, seq 1, length 64
16:28:23.604020 ARP, Request who-has 10.0.0.2 tell 10.0.0.1, length 28
16:28:23.604166 ARP, Request who-has 10.0.0.1 tell 10.0.0.2, length 28
16:28:23.604374 ARP, Reply 10.0.0.2 is-at 06:42:f5:d8:01:76 (oui Unknown), length 28
16:28:23.604493 ARP, Reply 10.0.0.1 is-at 7e:fa:a4:a6:e8:4c (oui Unknown), length 28
16:28:28.833849 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 1, length 64
16:28:28.834212 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 1, length 64
16:28:29.842744 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 2, length 64
16:28:29.842763 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 2, length 64
16:28:30.866779 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 3, length 64
16:28:30.866808 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 3, length 64
16:28:31.891031 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 4, length 64
16:28:31.891064 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 4, length 64
16:28:32.915433 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 5, length 64
16:28:32.915454 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 5, length 64
16:28:33.939423 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 6, length 64
16:28:33.939453 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 6, length 64
16:28:34.963219 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 7, length 64
16:28:34.963252 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 7, length 64
16:28:35.987270 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 8, length 64
16:28:35.987304 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 8, length 64
16:28:37.011777 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 9, length 64
16:28:37.011810 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 9, length 64
16:28:38.035982 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 10, length 64
16:28:38.036044 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 10, length 64
16:28:39.059800 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 11, length 64
16:28:39.059830 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 11, length 64
16:28:40.083098 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 12, length 64
16:28:40.083131 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 12, length 64
16:28:41.107682 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 13, length 64
16:28:41.107746 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 13, length 64
16:28:42.130829 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 14, length 64
16:28:42.130955 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 14, length 64
16:28:43.155951 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 15, length 64
16:28:43.156005 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 15, length 64
16:28:44.179729 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 16, length 64
16:28:44.179791 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 16, length 64
16:28:45.203320 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 17, length 64
16:28:45.203375 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 17, length 64
16:28:46.227456 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 18, length 64
16:28:46.227501 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 18, length 64
16:28:47.251599 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 19, length 64
16:28:47.251639 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 19, length 64
16:28:48.276233 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 20, length 64
16:28:48.276301 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 20, length 64
16:28:49.300066 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 2277, seq 21, length 64
16:28:49.300111 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 2277, seq 21, length 64
16:28:53.299991 ARP, Request who-has 10.0.0.1 tell 10.0.0.2, length 28
16:28:53.300163 ARP, Reply 10.0.0.1 is-at 7e:fa:a4:a6:e8:4c (oui Unknown), length 28
^C
52 packets captured
52 packets received by filter
0 packets dropped by kernel
```

Listens on the link between s1 and s2

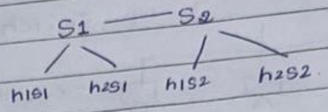
Captures all packets flowing between switches

Topology & Commands

5|1|26

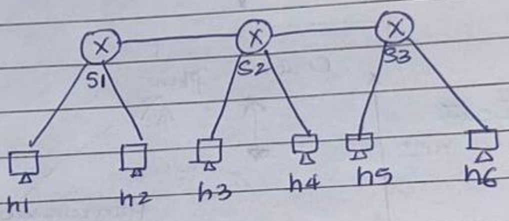
1. Sudo mn --topo linear, 2, 2
   2 - no of Switches (S1, S2)
   2 - no of hosts per Switch

Topology :-



2. Sudo mn --topo linear;3, 2 (Liniar)

   3 - no of Switches (S1, S2, S3)
   2 - hosts per switch

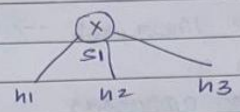


- Sudo mn :- Start mininet with root privilages.
- --topo :- This flag Specifies n/w topology
- linear :- topology where Switches are connected in a line.

depth = no of Switches level
fanout = no of children per Switch

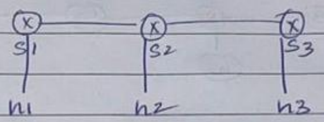
Linear = count Switch first
Single = count hosts directly

3. Sudo mn --topo single, 3.



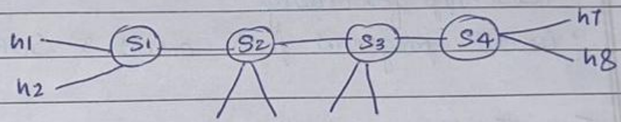
4. Sudo mn -topo linear, 3, 1

3 - Switches
1 - host per Switch



* 5. Custom Topology - 4 Switches, unequal hosts

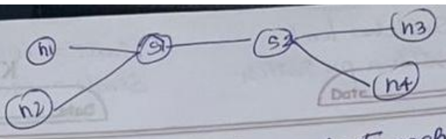
Eg.



→ 6. Terminal 1 :- ryu-manager ryu.app.simple_switch_13

ctl+Alt+F2 • Starts ryu controller
• Uses OpenFlow 1.3
• Learn MAC addresses
• Installs flow rules dynamically.

ctl+Alt+F1 -- controller = remote, ip = 127.0.0.1
↳ this connects Mininet Switches to Ryu instead of default controller.

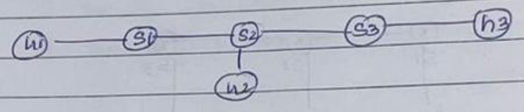


1. Linear Topology - 2 Switches, 2 hosts each

→ sudo mn --topo linear,2,2 --Controller=remote,ip=127.0.0.1

- Ryu learns MAC addresses
- Install flow rules on both Switches
- End-to-End Communication Enabled.

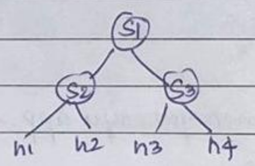
2. Linear Topology - 3 Switches, 1 hosts each.



→ sudo mn --topo linear,3,1 --controller=remote,ip=127.0.0.1

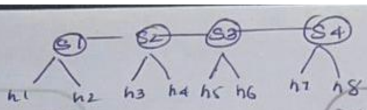
- Multi-hop forwording
- Controller programs paths across 3 Switches.

3. Tree Topology - Depth 2, Fanout 2



→ sudo mn --topo tree,2,2 --Controller-remote,ip=127.0.0.1

- Hierarchical forwording
- Ryu handles multiples branching paths
- Common data-Center style topology.



a. Linear - 4 Switches, 2 host each

→ sudo mn --topo linear,4,2 --controller=remote, ip=127.0.0.1, port=6653

- Scalability
- Flow installation across many Switches.

b. Using tcpdump to monitor Network flows in Mininet.

t1 → sudo mn --topo linear,2,1

h1 — s1 — s2 — h2

→ net # to check interfaces.

t2 → ctl+Alt+F2 → sudo tcpdump -i s1-eth2
- Listens on the link between S1 and S2
- Capture all packets flowing b/w switches.

To save → sudo tcpdump -i s1-eth2 -w ping.pcap